BY ORDER OF THE COMMANDER
99TH AIR BASE WING (ACC)

NELLIS AIR FORCE BASE
INSTRUCTION 15-129

23 MARCH 2022

Weather

WEATHER SUPPORT

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing website at www.e-Publishing.af.mil for downloading or ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: 57 OSS/OSW

Certified by: 57 OG/CC
(Colonel Scott C. Mills)

Supersedes: NELLISAFBI15-129, 31 Oct 2017

Pages: 49

This instruction establishes the responsibilities and procedures for providing and using weather services at Nellis AFB and it applies to all agencies described herein. It also implements and incorporates the requirements of Air Force Instruction (AFI) 15-128 ACCSUP, *Air Force Weather Roles and Responsibilities*, Air Force Manual 15-129 and *Air and Space Weather Operations*. It provides general information on weather services including weather observations and forecasts, weather warnings and advisories, dissemination of weather information, and reciprocal support. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See AFI 33-360, Publications and Forms Management, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the requestor's commander for non-tiered compliance items. Send comments and suggested improvements to this instruction on AF Form 847, *Recommendation for Change of Publication*, to 57 OSS/OSW, 6278 Depot Road, Building 805 Nellis Air Force Base, NV 89191. Ensure that all records created as a result of processes prescribed in this publication are maintained and disposed of in accordance with Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS).

SUMMARY OF CHANGES

This manual has been updated to incorporate changes to guiding publications and to remain in compliance with those publications. Notable updates include the weather watch, warning and advisory responsibilities of the 57th OSS Weather Flight and 25th Operational Weather Squadron

as well as the alert criteria for Nellis AFB. Lead time for severe wind events have been updated to increase lead time by an additional 30 minutes. Weather Flight operating hours were updated for 24/7 support. Severe Weather Conference call between 57Wg and 99th ABW has been established for events for predictable major impacts to base infrastructure and personnel.

Chapter 1—GENERAL INFORMATION **5**

1.1. General. 5

1.2. Services 5

1.3. Location and Hours of Operation. 5

Table 1.1. Weather Flight Contact Information. 5

Table 1.2. 25 OWS Contact Information. 6

1.4. Geographic Area of Responsibility. 6

1.5. Duty Priorities. 6

Table 1.3. Weather Flight Duty Priorities. 7

1.6. Forecasting Limitations 7

1.7. Dissemination of Weather Information. 7

1.8. Weather Station Identifiers. 8

1.9. Release of Weather Information. 8

1.10. Records Retention. 8

1.11. Changes to this Instruction. 8

Chapter 2—AIRFIELD WEATHER OPERATIONS **9**

2.1. General. 9

2.2. Primary Observation Equipment. 9

2.3. Additional Equipment. 9

2.4. Equipment Limitations. 10

2.5. Weather Observations and Limitations. 10

2.6. Augmentation of Primary Observation Equipment. 11

2.7. Meteorological Watch (METWATCH). 12

2.8. Continuous Weather Watch (CWW). 12

2.9. Basic Weather Watch (BWW). 12

2.10. Cooperative Weather Watch. 12

2.11. Pilot to METRO Service (PMSV). 13

2.12. PIREPs. 13

2.13. Weather Equipment Outages. 13

Table 2.1. Observation Phone Dissemination. 13
2.14. Alternate Operating Location (AOL). 14
Table 2.2. AOL Squadron Notification List. 15
2.15. Aircraft and Ground Mishaps. 15
2.16. Terminal Aerodrome Forecast (TAF). 16
Chapter 3—MISSION WEATHER OPERATIONS **17**
3.1. General. 17
3.2. Mission Planning Forecast (MPF). 17
3.3. Mission Execution Forecast (MEF). 17
3.4. Flight Weather Briefing (FWB). 17
3.5. Tactical Decision Aids (TDA). 18
3.6. Mission-Scale Meteorological Watch (MISSIONWATCH). 18
3.7. RED FLAG/GREEN FLAG-West. 18
3.8. USAFWS Weapons Integration/Advanced Integration (WI/AI). 19
3.9. Coronet Mission Briefings. 19
3.10. Supervisor of Flying (SOF) Briefing. 19
3.11. Out of Station Briefings. 19
Chapter 4—STAFF INTEGRATION **20**
4.1. General. 20
4.2. Stand-up Briefings. 20
Table 4.1. 4-Day Outlook Degradation Stop Light. 20
4.3. Airfield Operations Board (AOB). 21
4.4. Base Support Plan (BSP). 21
4.5. Instrument Refresher Course (IRC) Briefing. 21
4.6. Flight Information Publication (FLIP) Review. 21
4.7. Toxic Corridor, Chemical and Effective Downwind Messages. 21
Chapter 5—RESOURCE PROTECTION **22**
5.1. General. 22
5.2. Definition of Watch, Warning, Advisory (WWA) Terms. 22
5.3. Delineation of Weather Watch, Warning and Advisory Duties. 22
5.4. WWA Numbering. 22
5.5. Primary Dissemination. 22

Figure 5.1. Figure 5.1.Primary WWA Notification Flow. 23
5.6. Backup Dissemination. 23
5.7. Severe Weather Action Plan (SWAP). 23
5.8. Operational Reporting (OPREP). 24
Chapter 6—RECIPROCAL SUPPORT 25
6.1. General. 25
6.2. 99 ABW/CC will: 25
6.3. 99 ABW/CP will: 25
6.4. 99 CS will: 25
6.5. 57 OSS/OSAM will: 25
6.6. 99 OMRS/SGPB (Bioenvironmental Engineering Flight) will: 25
6.7. 57 WG/SE will: 26
6.8. 57 OG/CC will: 26
6.9. 57 OSS/OSA will: 26
6.10. 57 OSS/OSAA will: 26
6.11. 57 OSS/OSAR will: 26
6.12. 57 OSS/OSAT will: 26
6.13. Supervisor of Flying will: 27
6.14. 414 CTS, 549 CTS will: 27
6.15. 16 WPS, 17 WPS, 34 WPS, 66 WPS, 433 WPS, 64AGRS will: 27
6.16. 25 OWS will: 27
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 28
Attachment 2—LOCATION OF AN/FMQ-19 33
Attachment 3—SPECIAL OBSERVATION CRITERIA 34
Attachment 4—TAF SPECIFICATION AND AMENDMENT CRITERIA 37
Attachment 5—MISSION WEATHER THRESHOLDS / AIRCRAFT SENSITIVITIES 40
Attachment 6—MISSION EXECUTION FORECAST (MEF) AMENDMENT CRITERIA 43
Attachment 7—WEATHER WATCH, WARNING, AND ADVISORY (WWA) CRITERIA 46

Chapter 1

GENERAL INFORMATION

1.1. General. This document provides a consolidated source of weather support capabilities, procedures, and responsibilities at Nellis AFB, NV. It does not eliminate the need for specialized weather support to be included in base/command operational plans. Air Force Weather concepts and procedures are outlined in Air Force and MAJCOM publications.

1.2. Services . The 57th Operations Support Squadron Weather Flight (57 OSS/OSW) provides or arranges for weather support to the 57th Wing (57 WG), the 99th Air Base Wing (99 ABW), and tenant units. In addition, the weather flight issues observed and forecast weather watches, warnings and advisories for Nellis AFB with continuity of operations (COOP) support from the 25th Operational Weather Squadron at Davis-Monthan AFB, AZ. In concert with the 25 OWS, 57 OSS/OSW tailors products for criteria important to flying operations, base support agencies, and maintenance operations.

1.3. Location and Hours of Operation. The weather flight is located in Building 805, and is collocated with Airfield Management Operations. The weather flight will operate 24 hours a day, 7 days a week. Weather observations are automatically taken and disseminated through the weather flight's automated observation system; this continues even when the weather flight is closed.

1.3.1. The operating hours for the Airfield Weather Operations section are 24/7.

1.3.2. The Mission Weather Operations section is predominately located at the USAF Weapons School ops desk. During Red and Green Flag exercises, a forecaster will also be located at the respective squadron. Mission Weather Operations hours are established around the battle rhythm and flex based upon the mission requirements of Nellis AFB.

Table 1.1. Weather Flight Contact Information.

Weather Flight Commander	DSN 682-8707	Comm (702) 652-8707
Weather Flight Flight Chief	DSN 682-8730	Comm (702) 652-8730
Airfield Weather Operations	DSN 682-4744	Comm (702) 652-4744
Weapons School Mission Weather Operations	DSN 682-8868	Comm (702) 652-8868
Red Flag Mission Integration	DSN 682-2395	Comm (702) 652-2395
Green Flag Mission Integration	DSN 682-7030	Comm (702) 652-4262
Standby forecaster	DSN 682- 2446 / 2447 (Command Post, which will contact the standby forecaster)	Comm (702) 423-8185 (Mobile phone carried by standby forecaster)
Alternate Operating Location (see paragraph 04)	DSN 682-4551	Comm (702) 652-4551

Table 1.2. 25 OWS Contact Information.

Flight Weather Briefer	Voice DSN Fax DSN	228-6598/6599/6588 228-7361	Comm Fax	(520) 228-6598/6599/6588 (520) 228-7361
Senior Duty Officer	DSN	228-7655	Comm	(520) 228-7655
Ops Superintendent	DSN	228-2445	Comm	(520) 228-2445

1.4. Geographic Area of Responsibility. The area of responsibility for products and services provided by the weather flight is defined as the area located within a five nautical mile radius around the center of the Nellis AFB airfield complex. The weather flight will also conduct a meteorological watch (METWATCH) and a mission meteorological watch (MISSIONWATCH) for all areas and routes in which Nellis AFB flying units are conducting operations. These areas include, but are not limited to, the Nevada Test and Training Range (NTTR), the Utah Test and Training Range (UTTR), and Range 2508 located at the National Training Center (NTC) at Ft Irwin.

1.5. Duty Priorities. All AF weather units are required to develop clearly defined duty priorities. The weather flight duty priorities are listed in **Table 1.3**.

Table 1.3. Weather Flight Duty Priorities.

Priority	Duties
1	Perform Emergency War Order tasking's.
2	Evacuation of Weather Flight during emergencies.
3	Issue/Disseminate Weather Warnings.
4	Respond to aircraft and ground emergencies.
5	Respond to Pilot to Metro Service (PMSV) contacts. **Note:** This includes all contacts with airborne aircrews including phone patches, radio contacts, aircrew-generated computer messages, and contacts through satellite communications equipment and cell phones.
6	Provide weather information for the Supervisor of Flying (SOF).
7	Activate Severe Weather Action Procedures (SWAP).
8	Disseminate Urgent PIREPs.
9	Issue/Disseminate Weather Watches and Advisories.
10	Augment automated observing system.
11	Produce and disseminate TAF to include amendments.
12	Mission Execution Forecast Process -- Produce and Disseminate Forecasts to include amendments.
13	Perform MISSIONWATCH Activities.
14	Disseminate PIREPs.
15	Collaborate with 25 OWS.
16	Provide Briefings.
17	Accomplish Weather Functional Training.
18	Perform Administrative Tasks.

1.6. Forecasting Limitations

1.6.1. Limitations. The terrain features surrounding Nellis AFB and their effects on local weather patterns create forecasting and observing challenges. Therefore, forecasts beyond 30 hours will be used for planning purposes only.

1.6.2. Surrounding hills and mountains may block, enhance, or divert winds resulting in varying wind conditions across the airfield.

1.7. Dissemination of Weather Information. Timely and reliable dissemination of weather information is crucial for safety of flight and resource protection. The Weather Flight disseminates information in several ways.

1.7.1. AN/FMH-5, commonly known as the Joint Environmental Toolkit (JET) , is the primary means used by the weather flight and the 25 OWS to create and disseminate forecasts, observations, and weather warnings, watches, and advisories. The Air Traffic Control (ATC) Tower and the Nellis Air Traffic Control Facility (NATCF, referred to as Radar Approach Control or RAPCON at other bases) are equipped with Airfield Automation System (AFAS) monitors to view JET information; any computer connected to the base network can also view JET information.

1.7.1.1. If JET is inoperative the backup procedures listed in **paragraph 0** Will be followed.

1.7.2. Weather information can be accessed through NIPRNET. The weather flight SharePoint site displays real-time information from JET, the latest satellite image and radar data, and the most recent Mission Planning Forecast and Mission Execution Forecast. The site is reached using the link titled "Weather" on the Nellis AFB SharePoint at https://usaf.dps.mil/sites/57WG/57OG/57_Operations%20Support%20Squadron/OSW/default.aspx**.** Additional weather products such as hazard charts and regional weather data can be found on the 25 OWS page: https://25ows.us.af.mil/.

1.7.3. Nellis AFB agencies will be notified of weather watches, warnings, and advisories as detailed in **paragraph 5.5**.

1.8. Weather Station Identifiers. Forecasts and observations for Nellis AFB use the International Civil Aviation Organization (ICAO) identifier KLSV. The World Meteorological Organization (WMO) index number for KLSV is 72385.

1.9. Release of Weather Information. The weather flight will not provide weather information or support to non-DoD agencies or the public unless the request was coordinated with and approved by the 57th Wing Public Affairs Office (57 WG/PA).

1.10. Records Retention. All flight weather briefings, products produced by the weather flight, and mission-related contacts, regardless of format (i.e. electronic, verbal, or paper), will be documented and maintained IAW the AF Records Disposition Schedule Series 15.

1.11. Changes to this Instruction. Modifications to this instruction will be coordinated through the weather flight biennially, or as changes to weather support requirements dictate.

Chapter 2

AIRFIELD WEATHER OPERATIONS

2.1. General. The weather flight continually takes and disseminates routine and special weather observations throughout the day. The observation equipment operates in full automated mode except under the conditions described in **paragraph 0**, meaning most observations will be sent without input from weather personnel.

2.2. Primary Observation Equipment. The AN/FMQ-19 Fixed Based Weather Observing System (FBWOS) is used to take official weather observations at Nellis AFB. It interfaces directly with JET, automatically disseminating observations locally and long-line. As shown in **Attachment 2**, the AN/FMQ-19 has two sensor sites, one located near the touchdown point of each runway. The sensors measure the following elements:

2.2.1. Wind direction and velocity.

2.2.2. Cloud heights and coverage.

2.2.3. Visibility.

2.2.4. Present weather.

2.2.5. Runway Visual Range (RVR).

2.2.6. Atmospheric pressure (runway 21 site only).

2.2.7. Temperature (runway 21 site only).

2.2.8. Dew point temperature (from relative humidity – runway 21 site only).

2.2.9. Precipitation amount (runway 21 site only).

2.2.10. Lightning distance and bearing (runway 21 site only).

2.3. Additional Equipment. In addition to the AN/FMQ-19, the weather flight relies on the following equipment:

2.3.1. Gibson Ridge Radar Software. The weather flight displays and analyzes high-resolution weather radar data derived directly from national weather radars through this internet software program. The nearest weather radar is located on Nelson's Peak (approximately 30NM to the south of Nellis AFB) at an elevation of over 4,900 ft. This radar is controlled by the Las Vegas National Weather Service office. The weather flight also maintains a subscription with WeatherTap.com, a commercial internet weather information provider, as an alternate source for radar data.

2.3.2. Pilot to Metro Service (PMSV) Radio. The weather flight monitors frequency 323.9MHz. See **paragraph 2.11**for PMSV details.

2.3.3. Local Area Network (LAN). The LAN is vital to weather operations. The weather flight uses the LAN to access and disseminate forecasts, observations, watches, warnings, advisories, satellite imagery, radar data, and lightning strike information. The weather flight also relies on the LAN to obtain flight weather briefing products and other operational products from the 25 OWS.

2.4. Equipment Limitations. Currently known equipment limitations are listed below.

2.4.1. AN/FMQ-19 Limitations.

2.4.1.1. Cloud height and coverage are determined by laser beam ceilometers, which look at a small portion of the atmosphere directly above the sensor. A representative sky condition is calculated for the entire celestial dome using a time-averaged algorithm. The FMQ-19 measures cloud heights between 100 and 25,000 feet.

2.4.1.2. Visibility is determined at the sensor location so it may not be representative of the entire airfield.

2.4.1.3. The FMQ-19 may not detect all thunderstorm and lightning activity and it cannot detect hail, volcanic ash, cloud types, or tornadic activity.

2.4.1.4. The FMQ-19 may be slow to report accurate sky conditions, visibility, and present weather during periods of rapidly changing weather conditions because these elements are calculated using time-averaged algorithms.

2.4.2. PMSV is unreliable from:

2.4.2.1. 085° through 155° beyond 35NM at or below FL200.

2.4.2.2. 315° through 005° beyond 40NM at or below FL200.

2.4.2.3. 230° through 290° beyond 55NM at or below FL200.

2.4.3. Weather Radar. There are currently no known impacts to the local weather radar.

2.5. Weather Observations and Limitations.

2.5.1. The AN/FMQ-19 and JET are certified to operate automatically and all observations will be taken and disseminated in this manner unless the system fails partially (e.g. sensor failure) or completely (e.g. communication failure), or augmentation is required. The criteria for manual observations are described in **paragraph 0**.

2.5.2. Observing Locations/Observing Limitations**.** The official observing location for automated observations is the AN/FMQ-19 sensor location at the approach end of the active runway. The official observation location for manually observed elements is the compass rose painted approximately 50 feet southeast of AMOPS, building 805. Observing limitations for this location are:

2.5.2.1. Flight line facilities block the forecaster's view of the approach end of Runway 03L/R and the horizon from 190°-330°.

2.5.2.2. High intensity ramp lights diminish the ability to determine visibility at night.

2.5.2.3. Loud noises may prevent the forecaster from hearing thunder.

2.5.2.4. The Alternate Operating Location limitations are listed in **paragraph 0**.

2.5.3. Types of Observations. Two types of observations are disseminated to describe the weather at Nellis AFB: Aviation Routine Weather Report (METAR) and Non-routine (Special) Aviation Weather Report (SPECI). AFH 11-203V2, *Weather for Aircrews*, paragraph 2.5 gives a step-by-step example showing how to read METAR and SPECI observations.

2.5.3.1. METAR is a routine hourly observation as well as the primary observation code used throughout the world for reporting surface meteorological data. METAR reports contain information on wind, visibility, runway visual range, present weather and obscurations, sky condition, temperature, dew point, and altimeter setting. In addition, encoded and plain language information that elaborates on the weather conditions report may be appended to the report.

2.5.3.2. SPECI is an unscheduled observation transmitted when any of the special criteria have been observed. SPECI will contain all data elements found in a METAR report plus additional remarks that elaborate on data in the body of the report. All SPECI reports will be prepared and transmitted as soon as possible after the relevant criteria are observed. **Attachment 3** lists NAFB SPECI criteria.

2.6. Augmentation of Primary Observation Equipment. Modifying an observation generated by the FBWOS or manually taking an observation is known as augmentation. Augmentation is performed only by qualified weather flight personnel under specific circumstances. The two augmentation processes are *supplementin*g and *back-up*. A METAR which has been augmented by a qualified weather technician will be appended with AO2A.

2.6.1. Supplementing observations: Weather flight personnel will supplement the automated observation when weather is observed that is beyond the AN/FMQ-19's capability to detect. Weather technicians will perform a Basic Weather Watch (BWW) and be prepared to supplement observations when the airfield is open and the mandatory supplemented criteria are observed and/or forecast to occur within 1 hour. Weather personnel are required to log on to the JET system and be prepared to supplement whenever a watch or warning has been issued for tornadic activity regardless of airfield status. Weather personnel are not required to supplement during airfield closure hours for other criteria. The criteria for supplementing an observation are:

2.6.1.1. Tornado or funnel cloud.

2.6.1.2. Hail greater than or equal to 1/4".

2.6.1.3. Volcanic ash.

2.6.1.4. Ice pellets.

2.6.1.5. Freezing Precipitation.

2.6.1.6. Tower Visibility (4 SM or less and surface visibility differs by more than one reportable value).

2.6.2. Back-up observations: If the automated observation system is partially (due to individual sensor failure) or completely inoperative (due to system or communication failure) the weather flight will manually provide and disseminate weather data as necessary. The official observation point for manual observations is the compass rose painted approximately 50 feet southeast of AMOPS at building 805.

2.6.3. Recall/Augmentation procedures: During periods of airfield closure (e.g. holidays, down days, etc.), the standby forecaster will be available at all times to provide weather support to all airfield agencies. The dedicated standby cell phone and Mission Weather Operations numbers are listed in **Table 1.2** Upon notification of recall a forecaster will be ready to supplement airfield observations within 1 hour.

2.7. Meteorological Watch (METWATCH). METWATCH is the process of monitoring terrestrial and space weather in an area. The purpose of the METWATCH is to identify when and where observed conditions significantly diverge from forecasted conditions and to update or amend forecast products and notify supported agencies.

2.7.1. The weather flight conducts a METWATCH for Nellis AFB during flight operating hours and when recalled for WWA responsibilities as defined in section 5.3.

2.7.2. The 25 OWS provides a METWATCH for Nellis AFB for Weather Watch and Special Weather Statement support. They will also provide coverage when the weather flight is closed and recall standby weather flight personnel, as required.

2.8. Continuous Weather Watch (CWW). During a CWW, weather conditions will be continuously monitored and the observer will perform no other significant duties. At Nellis AFB the FBWOS performs an automatic CWW, though the weather flight may perform a CWW if locally determined to be more appropriate due to existing weather conditions.

2.9. Basic Weather Watch (BWW). The purpose of a BWW is to determine the need for a SPECI observation. During a BWW weather personnel will check weather conditions, at intervals not to exceed 20 minutes since the last observation or check, when any of the following conditions are observed to be occurring or forecast to occur within 1 hour:

2.9.1. Any change in weather conditions that may require a SPECI or LOCAL observation.

2.9.2. All supplemental and back up-criteria specified in section 2.6.

2.10. Cooperative Weather Watch. To ensure accuracy in weather observations, a Cooperative Weather Watch has been established between the weather flight and the 57 OSS Airfield Operations Flight (57 OSS/OSA). The primary concern of the Cooperative Weather Watch is tower visibility different than the reported surface visibility, local PIREPs, and any unreported weather conditions that could affect flight safety or location operations and resources.

2.10.1. The weather flight will provide initial visibility observation training and certification to control tower personnel. Use of AFAS and JET are part of this initial training. The weather flight will also assist with updating and validating tower visibility charts annually.

2.10.2. ATC personnel are able to assist the weather flight with resource protection and flight safety in several ways.

2.10.2.1. Control tower personnel who are certified to evaluate tower visibility will notify the weather flight or standby forecaster when: (1) ATC personnel disseminate a visibility report for the WF, (2) it meets the criteria for a tower visibility remark (less than 4 SM & differs by a reportable value from prevailing visibility), and (3) the observation crosses or matches an operationally significant ATC threshold.

2.10.2.2. Control tower personnel will report any occurrence of unreported weather conditions that could affect flight safety or be critical to the safety or efficiency of other local operations or resources.

2.10.2.3. ATC personnel will solicit PIREPs and relay all PIREPs received to the weather flight for dissemination within 5 minutes or as soon as possible.

2.11. Pilot to METRO Service (PMSV). The weather flight monitors 323.9 MHz to respond to requests for weather information from aircrews in flight or on the ground. The weather flight will solicit PIREPs from aircrews contacting the weather flight. PMSV is not available when the weather flight is closed. PMSV radio limitations are listed in **paragraph 0** and the PMSV radio outage contingency plan is described in **paragraph 0**.

2.12. PIREPs. All PIREPs received by the weather flight will be disseminated over JET system or through a backup method.

2.12.1. Any base agency receiving a PIREP will relay it within 5 minutes of receipt to the weather flight for dissemination.

2.13. Weather Equipment Outages. The weather flight has developed procedures to continue operations and mission support in the event of equipment or communication failure.

2.13.1. AN/FMQ-19 sensor or equipment failure:

2.13.1.1. Air Force Weather Web Services (AFWEBS) will be the primary alternative for lightning data. Commercial internet sources may be utilized as final option.

2.13.1.2. For sensor outages, other than lightning detection, weather flight personnel will switch to the inactive sensor site if possible (**paragraph 0** lists the sensors in each site) or use tactical equipment at the official manual observation location or AOL. Wind and pressure values from tactical equipment are reported as estimated. Any unrepresentative meteorological values will be reported as missing (M).

2.13.2. JET: AFWEBS is the primary source for alternative observation dissemination. Weather flight personnel will request observation transmission by another Air Force weather unit and verbally pass the observation to the agencies in **Table 2.1** Weather watches, warnings, and advisories will also be relayed to these agencies by the issuing organization (i.e. 25 OWS or the weather flight).

Table 2.1. Observation Phone Dissemination.

Order of Dissemination	DSN	Commercial
Tower/SOF	682-9612/2708	(702) 652-9612/2708
NATCF (RAPCON)	682-4222	(702) 652-4222
Command Post	682-2446/2447	(702) 652-2446/2447
25 OWS	228-6674/6673	(520) 228-6673/6674
AMOPS	682-4600/4601	(702) 652-4600/4601

2.13.3. PMSV radio: For outages expected to be less than one hour in length, ATC personnel will monitor the PMSV frequency and relay PIREPS and requests for weather information. If the outage is expected to last more than one hour, AMOPS will send a Notice to Airman (NOTAM). The NOTAM will state PMSV service is unavailable and aircraft with phone patch capabilities should contact the 25OWS for assistance and current weather at the phone numbers given in **Table 1.2**.

2.13.4. LAN: The weather flight will use telephones, cell phones, or commercial Wi-Fi to continue weather support.

2.13.4.1. If required to disseminate observations or weather warnings, advisories, and watches by phone, the responsible agency (either the weather flight or 25 OWS) will call the agencies listed in **Table 2.1**.

2.13.4.2. If the outage also includes base telephones. A runner will be employed as needed to pass critical weather information to the SOF and Tower. Ramp net radios or cell phones may also be used as a means to pass critical info immediately. As manpower and duties allow, runners will attempt to pass weather data to other primary agencies. Supported agencies may have to provide a runner to get weather information from the weather flight.

2.14. Alternate Operating Location (AOL). The weather flight AOL is on the 10th floor of building 2064 (Air Traffic Control Tower), DSN 682-9585, commercial (702) 652-9585. In the event of an evacuation the weather flight will relocate to the AOL. The weather flight will continue to provide airfield, mission weather, resource protection, and staff integration services once established at the AOL.

2.14.1. Operations at the AOL are subject to the following site-specific limitations:

2.14.1.1. No PMSV radio. See **paragraph 0** for the backup procedure.

2.14.2. Weather flight personnel will notify the agencies listed in **Table 2.1** that weather operations are relocating or have relocated to the AOL as soon as it is practical.

2.14.3. Until weather operations are relocated and re-established, the 25 OWS will perform METWATCH, MISSIONWATCH and take full responsibility of all watches, warnings and advisories (WWA).

2.14.4. If the FBWOS is not fully operational, weather flight personnel will take and transmit an observation within 15 minutes of arrival at the AOL. To ensure information is relayed promptly the observation will be disseminated by phone to the agencies listed in **Table 1** before normal dissemination procedures are followed.

2.14.5. The weather flight will notify the squadrons listed in **Table 2.2.** of AOL operations and of impacts to weather support.

Table 2.2. AOL Squadron Notification List.

	DSN: 682-XXXX Commercial: (702) 652-XXXX	
	Voice	**Fax**
USAF Weapons School		
16 WPS	-3377	No fax
17 WPS	-8801	No fax
34 WPS	-6046	No fax
66 WPS	-8397	No fax
433 WPS	-8613	No fax
53rd Wing		
88 TES	-4930	-2168
422 TES	-4997	No fax
57th Wing		
Thunderbirds	-4115	-4116
64 AGRS	-2238	-6229
57 ATSS	-6500	No fax
549 CTS	-4262	
24 TASS	-2126	No fax
*Red Flag Wx	-2395	-5084
*Green Flag Wx	-7030	
Tenant Units		
66 RQS	-6692	No fax

2.15. Aircraft and Ground Mishaps. When notified of any aircraft or ground mishap (weather-related or not) requiring OPREP-3 or local reporting, the weather flight will contact the 25 OWS and collaborate to ensure applicable and available terrestrial and space weather data is archived.

2.15.1. The weather flight will coordinate with other relevant units (e.g. TACC for an AMC gained mission, nearest weather unit, etc) to ensure required data is saved.

2.15.2. The weather flight, 25 OWS, and other relevant units will archive data for the region of operations from 12 hours before the time of the event through 6 hours after the event.

2.16. Terminal Aerodrome Forecast (TAF). A TAF is an aviation forecast for a specific aerodrome providing meteorological information for flight planning and command and control. Unless otherwise specified, elements in the TAF apply to the area within a 5 statute mile radius of the center of the aerodrome. TAFs are disseminated as coded weather bulletins and AFH 11-203V2, *Weather for Aircrews*, paragraph 3.1 gives a step-by-step example showing how to read TAFs.

2.16.1. The TAF will be amended when conditions not in the forecast occur and are expected to last at least 30 minutes, or if forecast conditions do not occur by the specified hour and are not expected to occur within the next 30 minutes. TAF amendment criteria are listed in **Attachment 4**.

2.16.2. The Nellis TAF is issued every 8 hours, at 07Z, 15Z, and 23Z, and provides a 30 hour forecast. A TAF is normally not issued if the airfield will be closed for more than 8 hours.

2.16.3. The TAF will contain a LAST NO AMDS AFT D1D1H1H1 NEXT D2D2H2H2 when the airfield will be closed. Times are rounded to the nearest hour UTC. This remark is used when the next TAF will not be issued at its scheduled time, i.e. the airfield will be closing for more than 8 hours. $D_1D_1H_1H_1$ is the day and time after which the TAF will no longer be amended and $D_2D_2H_2H_2$ is the day and time the next TAF will be issued.

Chapter 3

MISSION WEATHER OPERATIONS

3.1. General. The weather flight provides mission-tailored weather services, METWATCH, and MISSIONWATCH.

3.2. Mission Planning Forecast (MPF). The Nellis MPF is produced by the weather flight for mission planners and contains a two day weather outlook for the NTTR. This non-amendable forecast is issued no later than 1600L on days the weather flight is open and the NTTR is in use. The MPF is issued on Thursday will be for Friday and Monday, the MPF issued on Friday will be for Monday and Tuesday. The MPF is posted to the weather flight SharePoint site, linked in **paragraph 1.7.2**.

3.3. Mission Execution Forecast (MEF). The MEF is a customized weather product providing terrestrial and space weather data and forecasts for a specific mission or set of missions that integrates aerospace weather with the customer's tactics, weapon systems, environmental sensitivities of equipment, and other operational requirements. The weather flight produces MEFs tailored to the types of aircraft and missions scheduled for the day. The issue time of the morning MEF depends on first take-off and may vary based on the flying schedule and weather conditions. The afternoon and evening MEFs are issued at 1200L and 1800L as necessary, Monday – Friday. MEFs are valid for 8 hours; weather flight personnel will update the current MEF if any of the criteria listed in **Attachment 6** occur.

3.3.1. MEFs will provide the following information:

3.3.1.1. Takeoff and landing data.

3.3.1.2. Range weather forecasts for the areas of operation.

3.3.1.3. Solar, lunar, and space weather data.

3.3.1.4. Forecasts for alternate airfields.

3.3.1.5. Space weather impacts to HF and UHF communications, forecast GPS error, and forecast high altitude radiation dosage.

3.3.1.6. The MEF is posted to the weather flight SharePoint site, linked in **paragraph 1.7.2** The format, timing, and dissemination of the MEF are determined by the requirements of supported units. The weather flight coordinates with each unit to determine the content and format of the MEF, ensuring it contains decision-grade information applicable to the mission.

3.4. Flight Weather Briefing (FWB). FWBs support aircrews requiring enroute weather when departing Nellis AFB and are produced by the weather flight as a DD Form 175-1 *Flight Weather Brief* or as a verbal briefing. If requested, FWBs can be posted to the weather flight SharePoint site, linked in **paragraph 1.7.2**.

3.4.1. The weather flight will provide FWB support to transient aircrews IAW duty priorities (see **Table 1Error**! No text of specified style in document.**-1**). If unable to assist the aircrew within 30 minutes the weather flight will either arrange for the 25 OWS to generate and deliver the FWB or provide contact information for the 25 OWS so the aircrew can arrange for a flight

weather briefing in the mission planning room. Under no circumstances will the weather flight deny assistance to an aircrew requesting a FWB.

3.4.2. AMC-gained missions: The weather flight will update takeoff weather data and notify Tanker Airlift Control Center (TACC) Global Mobility Weather Operations (XOW) if the update includes mission-limiting criteria IAW AFMAN 15-129 Paragraph 4.7.1.

3.4.3. ACC Air Operations Squadron (AOS) controlled missions: The weather flight will brief aircrews using ACC/AOC Controlling Mission Execution Forecast (CMEF). Any deviations from CMEFs will be coordinated with ACC/AOS.

3.4.4. As noted in **paragraph 1.10** all FWBs will be documented and maintained. The following items will be retained as part of the record:

3.4.4.1. Briefing time.

3.4.4.2. Briefer initials.

3.4.4.3. Aircrew call sign or mission number.

3.4.4.4. All information relayed to aircrew.

3.4.4.5. Void time, if applicable.

3.5. Tactical Decision Aids (TDA). Mission-specific electro-optical/infrared forecasts will be provided by the weather flight on request. Tactical Acquisition Weapons Software (TAWS), a DoD-certified application, is used by the weather flight to create TDAs for specific electro-optical weapons and sensor systems. The standard TDA produced by the weather flight includes maximum target detection range, lock-on range, and a graph of target and background temperatures over time. TDAs are posted to the weather flight SharePoint site (linked in **paragraph 1.7.2**) or will be e-mailed, if requested.

3.6. Mission-Scale Meteorological Watch (MISSIONWATCH). MISSIONWATCH is the process for monitoring terrestrial or space weather for predefined mission-limiting factors. The weather flight conducts a MISSIONWATCH tailored to the missions of the day and updates supported units and the SOF on changes to mission-critical conditions. The weather flight also updates the current MEF and ensures the TAF accurately reflects current and forecasted conditions. Transient aircrews are the exception to this policy.

3.7. RED FLAG/GREEN FLAG-West. Forecasters in support of RED FLAG or GREEN FLAG-West are responsible for all aspects of weather support to TDY aircrews and other participating personnel. Weather support includes but is not limited to mission weather briefs, weather updates, warnings, watches, and advisories and redeployment briefs after ENDEX. Weather support will be provided by a combination of Nellis and TDY weather personnel. In instances where no US forces are participating, or TDY weather personnel do no participate, the Nellis weather flight will provide or arrange all weather support with coordination through ACC/A3W. The Nellis weather flight will serve as the point of contact for TDY weather personnel and assist with the stand-up of the exercise weather area. In addition, the weather flight will provide familiarization training on weather patterns and forecasting challenges for the local area and required MOAs.

3.8. USAFWS Weapons Integration/Advanced Integration (WI/AI). The final phase of the Weapons School is the Advanced Integration. Weather flight personnel will provide weather support for the AI phase including coordination with the AI Project Officer on required briefings.

3.9. Coronet Mission Briefings. Weather flight personnel will coordinate briefing requirements with the Deployed Controlling Officer and brief aircrews as described in **paragraph 3.4.3**.

3.10. Supervisor of Flying (SOF) Briefing. One hour prior to the first scheduled flight the weather flight will provide the SOF a weather briefing in person or over the phone. The briefing covers current and forecasted conditions at the airfield and all military operating areas scheduled for use. As part of the MISSIONWATCH, the weather flight will alert the SOF to all mission-impacting changes to the weather. All briefings and updates will be documented and maintained as described in **paragraph 1.7.2**.

3.11. Out of Station Briefings. A supported unit may request weather flight participation in a mission briefing. Weather flight personnel will prepare and present a weather briefing tailored to the specific needs of the mission.

Chapter 4

STAFF INTEGRATION

4.1. General. Staff weather services are generally provided by the officers and senior NCOs of the weather flight and are tailored to the needs of the supported unit. The weather flight staff will provide or arrange for all weather support or information required by supported organizations. This includes, but is not limited to, climatological, historical, and planning weather data or studies, as well as astronomical and tidal data.

4.2. Stand-up Briefings. The weather flight will provide slides and/or a briefer for any Nellis Wing, Group or Squadron stand-up briefing upon request. A 4-day weather outlook will be posted to the Nellis SharePoint site at the following location https://usaf.dps.mil/sites/57WG/57OG/57_Operations%20Support%20Squadron/OSW/default.aspx.

4.2.1. The Nellis 4-Day Outlook is a non-amendable planning product produced Sunday through Thursday. **Table 4.1** indicates color-coded degradation markers.

Table 4.1. 4-Day Outlook Degradation Stop Light.

Operation	Favorable (Green)	Marginal (Yellow)	Unfavorable (Red)
Fixed Wing	Wind <35kts	Wind 35-49kts	Winds >= 50kts
	Ceiling >2000ft	Ceiling 1000-2000ft	Ceiling <1000ft
	Visibility >3200m	Visibility 1600-3200m	Visibility <1600m
			Thunderstorms
Rotary Wing	Wind <30kts	Wind 30-44kts	Wind >=45kts
	Ceiling >= 3000ft	Ceiling 1000-2999ft	Ceiling <1000ft
	Visibility >=4800m	Visibility 1601-4800m	Visibility <=1600m
			Thunderstorms
Ground Operations	No Precipitation	Light to Moderate Precipitation	Heavy Precipitation
	Wind <35kts	Wind 35-49kts	Wind >=50kts
	Temperature 32-85F	Temperature 20-32F Temperature 86-95F	Temperature <20F Temperature >95F

Illumination		Lunar Illumination>25% and Lunar Elevation 0-30°	Lunar Illumination <25% or Lunar Elevation <0°

4.3. Airfield Operations Board (AOB). A member of the weather flight staff will participate as a member of the AOB as directed in AFI 13-204V3, *Airfield Operations Procedures and Programs*.

4.4. Base Support Plan (BSP). The weather flight will designate a primary and an alternate BSP POC. These individuals are de facto members of the Base Support Planning Committee and are responsible for the currency and accuracy of the BSP weather chapter. They will collaborate with and provide support to other members of the Base Support Planning Committee, update the 57 OSS/CC on BSP issues, and maintain a continuity book documenting the discussions and rationale behind the BSP weather chapter. At least one POC will attend all BSP meetings.

4.5. Instrument Refresher Course (IRC) Briefing. The IRC briefing is a mandatory class for all personnel on flying status. The IRC weather brief is a refresher on local weather, weather hazards, weather equipment, and online tools and data available from the weather flight.

4.6. Flight Information Publication (FLIP) Review. The weather flight will review each new FLIP edition applicable to Nellis AFB immediately upon distribution to verify the accuracy of weather support information and will request changes through AMOPS. In addition, airfield minima will be checked to ensure SPECI observation criteria are correct. The 25 OWS will be notified of any modifications to SPECI criteria.

4.7. Toxic Corridor, Chemical and Effective Downwind Messages. IAW NAFB 10-2, *Comprehensive Emergency Management Plan*, upon notification of any incident that involves a toxic spill or base emergency, the weather flight will provide appropriate weather data for toxic corridor calculations performed by 99 CES/CEX. Updates will be provided as necessary.

4.7.1. Chemical Downwind Messages (CDM) and Effective Downwind Messages (EDM) are available upon request.

Chapter 5

RESOURCE PROTECTION

5.1. General. The 57 OSS weather flight provides resource protection through forecasts, METWATCH, and weather alerts providing enough advance notice to allow units to prepare.

5.2. Definition of Watch, Warning, Advisory (WWA) Terms.

5.2.1. Severe weather: Any weather condition that poses a hazard to property or life.

5.2.2. Weather watch: Issued when the potential exists for severe weather. Organizations and personnel should consider taking protective measures.

5.2.3. Weather warning: Issued when severe weather is imminent or occurring. Organizations and personnel must take immediate protective actions.

5.2.4. Weather advisory: Issued when non-severe weather conditions impacting operations are occurring or are forecast to occur.

5.2.5. Desired lead time (DLT): The total amount of time required to disseminate a forecast WWA from the issuing organization through the local dissemination tree to the supported agency plus the amount of advance notice a supported agency requires before the onset of the weather. For example, if the DLT is 30 minutes and it takes 5 minutes for the WWA to reach the supported agency, then that agency will have 25 minutes advance notice. WWA criteria are listed in **attachment 8**.

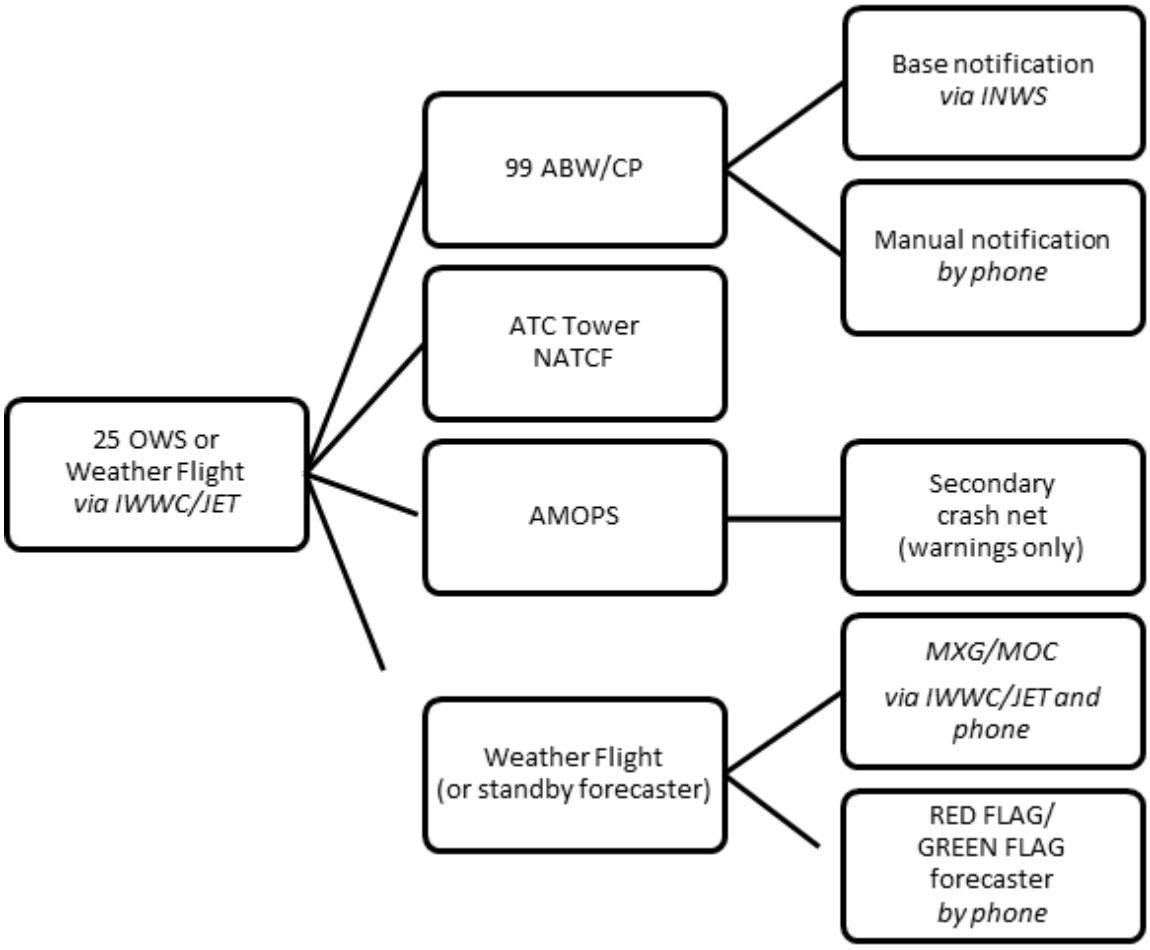
5.2.6. Special Weather Statement (SWS): Advance notice of widespread hazardous weather conditions with the potential to affect military installation in the specified geographic area. They are normally issued 48-72 hours in advance.

5.3. Delineation of Weather Watch, Warning and Advisory Duties. The 57 OSS weather flight is primarily responsible for issuing all forecast warnings, observed warnings, forecast lightning watches, forecast advisories and observed advisories;

5.4. WWA Numbering. Weather watches, warnings, and advisories are numbered consecutively by type. Each WWA is first identified by the type of message (watch, warning, or advisory) followed by an identification number using the format MM-XXX. MM is the two digit month of issue and XXX is the sequence number. For example, the message "Weather Warning 02-005" means the message was issued in February and was the fifth warning issued that month. The message "Weather Advisory 12-013" means the message was issued in December and was the thirteenth advisory issued that month.

5.5. Primary Dissemination. WWAs are disseminated through JET using the automated Integrated Weather Warning Capability (IWWC). The WWA is displayed on the AFAS monitors in the ATC Tower, NATCF and in JET at the weather flight. IWWC transmits an automated message via telephone to the command post and AMOPS. The command post uses the Nellis AFB Installation Notification Warning System (INWS) to send warnings, as required. AMOPS activates the secondary crash net IAW CEMP 10-2 for situations in **paragraph 6.10.1**. **Figure 5.1**illustrates this notification flow.

Figure 5.1. Primary WWA Notification Flow.



5.6. Backup Dissemination. If JET is inoperative, the organization issuing the WWA (either the 25 OWS or the weather flight) will contact the agencies listed in Table by phone. The contacted agencies will follow their backup notification procedures as required.

5.7. Severe Weather Action Plan (SWAP). The SWAP defines the events, personnel requirements, and operating procedures required to mitigate the threat of severe weather.

5.7.1. 57 OSS/OSW will: initiate a severe weather conference call through 99 ABW/CP when required. This conference call will be used to assess potential widespread severe weather hazards to Nellis AFB and/or the local area. An initial conference call should take place 24-48 hours in advance of the event to allow for maximum planning, when possible.

5.7.2. Conference call attendees will include at a minimum: 99ABW/CC, 57WG/CC, 57WG/SE, 99 MSG/CC, 99 CES/CEX, 57OSS/OSW.

5.7.2.1. Potential weather hazards for conference call include: Widespread and prolonged periods of wind gusting or sustained greater than or equal to 50kts, any frozen precipitation amounting in snow or ice accumulation, any flooding which could cause significant damage to base infrastructure, widespread organized severe thunderstorms with hail or wind greater than 50kts, or Tornado potential.

5.7.3. At a minimum the SWAP will be activated whenever watches or warnings are issued for the following conditions:

5.7.3.1. Tornado.

5.7.3.2. Severe thunderstorms (surface winds ≥ 50 kts associated with thunderstorms or hail ≥ 3/4 inch diameter).

5.7.3.3. Hail ≥ ¼ inch diameter.

5.7.3.4. Damaging winds (surface winds ≥ 50 kts not associated with thunderstorms).

5.7.3.5. Freezing precipitation.

5.7.4. The severe weather team consists of a team lead (one of the officers or NCOICs of the weather flight) and any weather flight personnel recalled by the team lead as required to efficiently handle weather conditions and increased workload.

5.7.4.1. SWAP personnel will perform enhanced METWATCH and MISSIONWATCH, increase surveillance of local and regional conditions and weather radar.

5.7.4.2. The SWAP leader, in addition to the duties in 5.7.3.1, will perform Operational Reporting (OPREP) duties (see **paragraph 5.8**).

5.7.4.3. SWAP will remain in effect until the expiration or cancellation of the watches and warnings responsible for SWAP activation.

5.8. Operational Reporting (OPREP). If requested by 99 ABW/CP, the weather flight commander, NCOIC, or the lead during SWAP activation will collaborate with the 25 OWS to provide weather data for the OPREP-3 report.

5.8.1. All weather information provided for a weather-related OPREP-3 report will also be sent to ACC/A3W and the 25 OWS.

Chapter 6

RECIPROCAL SUPPORT

6.1. General. The agencies listed in this chapter will provide services as described below.

6.2. 99 ABW/CC will: Chair a review of installation severe weather preparedness, capabilities, requirements, and procedures annually IAW AFI 10-229, *Responding To Severe Weather Events*.

6.3. 99 ABW/CP will:

6.3.1. Disseminate weather watches and warnings via INWS.

6.3.2. Confirm weather updates via JET IWWC.

6.3.3. Notify the weather flight of any aircraft or ground mishap (weather-related or not) requiring OPREP-3 or local reporting.

6.3.4. Provide a copy of OPREPs containing weather data to the weather flight.

6.3.5. Relay PIREPs or reports of significant weather (including turbulence and icing) to the weather flight.

6.3.6. Notify the weather flight of events that require weather support.

6.4. 99 CS will:

6.4.1. Provide support to the weather flight as directed by coordinated Service Level Agreements and Memorandums of Agreement.

6.4.2. Notify the weather flight at least 24 hours in advance of scheduled LAN or Internet outages. Notifications from the 99 CS will be distributed to the 57 OSS/OSW via the 57 OSS/OSW Organizational Email account.

6.5. 57 OSS/OSAM will:

6.5.1. Notify the weather flight of scheduled outages to the radio tower or equipment related to the weather flight's PMSV radio.

6.5.2. Inform the weather flight about alterations to the radio tower or equipment related to the weather flight's PMSV radio as alterations may affect the range of PMSV service.

6.5.3. Maintain weather equipment technical orders (TOs) and make them available to the weather flight upon request.

6.6. 99 OMRS/SGPB (Bioenvironmental Engineering Flight) will:

6.6.1. Provide the base populace the Wet Bulb Globe Temperature, Heat Stress Index, Wind Chill, and Frostbite Risk Level during normal duty hours (0730-1630 Monday-Friday excluding holidays, Air Force Family Days, and ACC Family Days).

6.6.2. Assist the weather flight in validating and confirming appropriate fighter induced thermal stress (FITS) index levels and can be reached at DSN 653-3316.

6.6.3. Upon request, provide base populace and commanders with health risk assessments and recommendations based on Wet Bulb Globe Temperature, Heat Stress Index, Wind Chill, Frostbite Risk Level, and FITS Index Level.

6.7. 57 WG/SE will:

6.7.1. Notify the weather flight, as soon as possible, when any mishap occurs requiring a weather data save.

6.7.2. Notify the weather flight when a weather presentation is required for flight safety meetings.

6.8. 57 OG/CC will:

6.8.1. Establish operational weather support requirements and procedures.

6.8.2. Notify the weather flight of all changes in mission weather support.

6.8.3. Coordinate SOF, IRC, and seasonal awareness briefings with the weather flight.

6.9. 57 OSS/OSA will:

6.9.1. Provide a workspace with a telephone, LAN access, and 24-hour physical access to the 10th floor of building 2064 (ATC Tower) as the weather flight Alternate Operating Location.

6.10. 57 OSS/OSAA will:

6.10.1. Provide the weather flight updated DoD Flight Information Publications (FLIPs).

6.10.2. Notify the weather flight of permanent or temporary changes to airfield operating hours.

6.10.3. Coordinate with the weather flight before switching Building 805 between commercial and generator power.

6.10.4. Disseminate all warnings for lightning within 5 NM over the Tower Net.

6.11. 57 OSS/OSAR will:

6.11.1. Relay PIREPs to the weather flight within 5 minutes of receipt.

6.11.2. Monitor JET weather data on AFAS and report JET outages or restoration of services to the weather flight (or standby forecaster).

6.12. 57 OSS/OSAT will:

6.12.1. Relay runway and wind sensor changes to the weather flight.

6.12.2. Relay PIREPs to the weather flight within 5 minutes of receipt.

6.12.3. Monitor JET weather data on AFAS and report JET outages or restoration of services to the weather flight (or standby forecaster).

6.12.4. Participate in the Cooperative Weather Watch. See **paragraph 2.10**.

6.12.5. Perform a PMSV radio check upon request.

6.12.6. Monitor PMSV frequency as time permits if the weather flight's PMSV radio is inoperative or the weather flight evacuated building 805. See **paragraph 0**.

6.12.7. Provide ATC Tower indoctrination and orientation training for newly-assigned weather personnel.

6.12.8. In the event of an aircraft divert after dark, will turn on the runway lights as far in advance of the divert as possible in order to allow the FMQ-19 RVR sensor enough time to accurately measure runway visual range.

6.13. Supervisor of Flying will:

6.13.1. Contact the weather flight for a weather briefing prior to the first scheduled takeoff IAW AFI 11-418 Nellis AFB Supplement, Operations Supervision, paragraph 4.3.13.1..

6.13.2. Relay PIREPs to the weather flight within 5 minutes of receipt.

6.14. 414 CTS, 549 CTS will:

6.14.1. Ensure deployed aircrews participating in RED FLAG and GREEN FLAG know to contact their assigned TDY forecasters for weather support.

6.14.2. Encourage deployed aircrews to submit PIREPs upon encountering significant weather (including turbulence and icing) for dissemination by their assigned weather personnel.

6.14.3. Coordinate TDA and NVG support at least 4 hours prior to brief time.

6.15. 16 WPS, 17 WPS, 34 WPS, 66 WPS, 433 WPS, 64AGRS will:

6.15.1. Coordinate TDA and NVG support at least 4 hours prior to brief time.

6.15.2. Submit a PIREP upon encountering significant weather (including turbulence and icing).

6.16. 25 OWS will:

6.16.1. Conduct METWATCH, WWA issuance, TAF issuance, flight weather brief creation and/or dissemination of observations under emergency COOP procedures until 57 OSS/OSW establishes operations at AOL.

TODD R. DYER, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFPD 15-1, *Weather Operations*, 14 Nov 2019

AFI 11-418 Nellis AFB Supplement, *Operations Supervision*, 2 Jun 2016

AFI 15-128_ACCSUP, *Air Force Weather Roles and Responsibilities*, 16 Mar 2020

AFMAN 15-111, *Surface Weather Observations*, 12 Mar 2019

AFMAN 15-124, *Meteorological Codes*, 16 Jan 2019

AFMAN 15-129, *Air and Space Weather Operations,* 9 Jul 2020

AFH 11-203V2, *Weather for Aircrews*, 13 Aug 2015

AFPAM 48-151, *Thermal Injury*, 7 Apr 2016

NAFB/CAFB/NTTR, Plan 10-2, *Installation Emergency Management Plan*, 1 Dec 2017

Prescribed Forms

DD Form 175-1, *Flight Weather Briefing*

Abbreviations and Acronyms

AFAS—Airfield Automation System

AFT—After

AFWEBS—Air Force Weather Web Services

AMDS—Amendments

AMOPS—Airfield Management Operations

AMOS—Automated Observing System

AOB—Airfield Operations Board

AOL—Alternate Operating Location

AOS—Air Operations Squadron

ATC—Air Traffic Control

BSP—Base Support Plan

BWW—Basic Weather Watch

CDM—Chemical Downwind Message

CMEF—Controlling Mission Execution Forecast

CP—Command Post

CWW—Continuous Weather Watch

DLT—Desired Lead Time

EDM—Effective Downwind Message

FBWOS—Fixed Based Weather Observing System

FLIP—Flight Information Publication

FWB—Flight Weather Briefing

ICAO—International Civil Aviation Organization

INWS—Installation Notification Warning System

IRC—Instrument Refresher Course

IWWC—Integrated Weather Warning Capability

JET—Joint Environmental Toolkit

LAN—Local Area Network

MEF—Mission Execution Forecast

METAR—Aviation Routine Weather Report

METWATCH—Meteorological Watch

MISSIONWATCH—Mission Watch

MOA—Military Operating Area

MOAF—Military Operating Area Forecast

MPF—Mission Planning Forecast

NATCF—Nellis Air Traffic Control Facility

NOTAM—Notice to Airmen

NTC—National Training Center

NTTR—Nevada Test and Training Range

NVG—Night Vision Goggles

NWS—National Weather Service

OPREP—Operational Report

OWS—National Weather Service

PIREP—Pilot Report

PMSV—Pilot-to-Metro Service

RAPCON—Radar Approach Control

RVR—Runway Visual Range

SOF—Supervisor of Flying

SPECI—Special Aviation Report

SWAP—Severe Weather Action Plan

SWS—Special Weather Statement

TACC—Tanker Airlift Control Center

TAF—Terminal Aerodrome Forecast

TAWS—Tactical Acquisition Weapons Software

TDA—Tactical Decision Aid

TO—Technical Order

USAFWS—USAF Weapons School

UTTR—Utah Test and Training Range

WI—Weapons Integration

WF—Weather Flight

WMO—World Meteorological Organization

WWA—Watch, Warning, Advisory

Terms

Alternate Operating Location (AOL)—The location to which the weather flight will move in the event that Bldg 305 is evacuated.

Climatology—Historical records of weather conditions measured or observed at a specific location. Climatology is useful in planning operations beyond 5 to 7 days. It usually describes the average (or mean) conditions such as high and low temperatures and extremes.

Continuous Weather Watch (CWW)—Continuous monitoring of weather conditions by a member of the weather flight during which no other significant duties are performed. Both routine (METAR) and special (SPECI) weather observations will be taken and disseminated appropriate due to existing meteorological conditions.

Desired Lead Time (DLT)—The total amount of time required to disseminate a forecast WWA from the issuing organization through the local dissemination tree to the supported agency plus the amount of advance notice a supported agency requires before the onset of the weather.

Eyes Forward—One of the functions the weather flight performs for the OWS in which weather flight forecasters relay meteorological information not contained in coded observations to the OWS. It is an integral part of the METWATCH.

Fixed Based Weather Observing System (FBWOS)—An accredited observing system with the capability to automatically collect and disseminate observations.

Installation Data Page—A document used by the 25 OWS defining the specific environmental support requirements, technical data, reference material, and contact information for Nellis AFB

International Civil Aviation Organization (ICAO)—A United Nations organization specializing in matters dealing with international aviation and navigation.

Joint Environmental Toolkit (JET)—The primary means used by the weather flight and the 25 OWS to create and disseminate forecasts, observations, and weather warnings, watches, and advisories.

METAR—METAR is a routine scheduled observation as well as the primary observation code used by the United States to satisfy requirements for reporting surface meteorological data.

Meteorological Watch (METWATCH)—The process of monitoring terrestrial and space weather in an area. The purpose of the METWATCH is to identify when and where observed conditions significantly diverge from forecasted conditions and to update or amend forecast products and notify supported agencies.

Military Operating Area (MOA)—A specific area in which military operations are occurring.

Military Operating Area Forecast (MOAF)—A forecast for terrestrial or space weather for a specific area in which military operations are occurring

Mission Execution Forecast (MEF)—Products generated by weather organizations that are focused on execution of aviation, space, ground (Operations Group, Mission Support Group or other organizational operations) or maritime operations.

Mission Planning Forecast (MPF)—A two-day weather non-amendable forecast for the NTTR, issued no later than 0500L on days the weather flight is open and the NTTR is in use.

Mission Watch (MISSIONWATCH)—The monitoring of aerospace weather for a specific mission (i.e., ground, air or space) and informing supported agencies when unforecast mission-limiting phenomena could affect operations.

Nellis Air Traffic Control Facility (NATCF)—The Radar Approach Control, or RAPCON, for Nellis AFB.

Notice to Airmen (NOTAM)—A timely notice containing information concerning the establishment, condition, or change in any aeronautical facility, service, procedures, or hazards, essential to personnel concerned with flight operations.

Operational Weather Squadron (OWS)—An organization responsible for providing regional weather support by producing operational-level weather forecast products and services. Nellis AFB is supported by the 25 OWS at Davis-Monthan AFB, AZ.

Pilot Report (PIREP)—A report of in-flight weather provided by an aircraft crewmember.

Pilot-to-Metro Service (PMSV)—A means for aircrews to request weather information and receive updates to flight weather briefings. Aircrew can also transmit PIREPs which are then disseminated worldwide.

Prevailing Visibility—The visibility considered to be representative of the visibility conditions at the official observing point.

Runway Visual Range (RVR)—The distance over which a pilot of an aircraft on the centerline of the runway can see the runway surface markings or its center line.

Severe Weather—Any weather condition that poses a hazard to property or life.

Severe Weather Action Procedures (SWAP)—Defines the events, personnel requirements, and operating procedures required to mitigate the threat of severe weather.

Space Weather—Describes the conditions in space that affect the near-Earth space environment, satellites, and ground based systems. Space weather forecasts include impacts to radio communications and GPS systems.

SPECI—An unscheduled observation taken when special weather criteria are observed.

Special Weather Statement (SWS)—An OWS notice to supported customers of meteorological effects which could impact future operations. This notice is for situational awareness purposes only and does not require action by supported customers.

Tactical Acquisition Weapons Software (TAWS)—Used by the weather flight to create TDAs for specific electro-optical weapons and sensor systems. The standard TDA produced by the weather flight includes maximum target detection range, lock-on range, and a graph of target and background temperatures over time.

Tactical Decision Aid (TDA)—A product showing the predicted maximum detection or lock-on range of air-to-ground electro-optical weapons and sensor systems.

Terminal Aerodrome Forecast (TAF)—An aviation forecast for a specific aerodrome providing meteorological information for flight planning and command and control, disseminated as coded weather bulletins.

Weather Advisory—Issued when non-severe weather conditions impacting operations are occurring or are forecast to occur.

Weather Products—Products generated by weather personnel for weather personnel or for non-weather personnel to use for planning purposes, situational awareness, and MEF generation.

Weather Warning—Issued when severe weather is imminent or occurring. Organizations and personnel must take immediate protective actions.

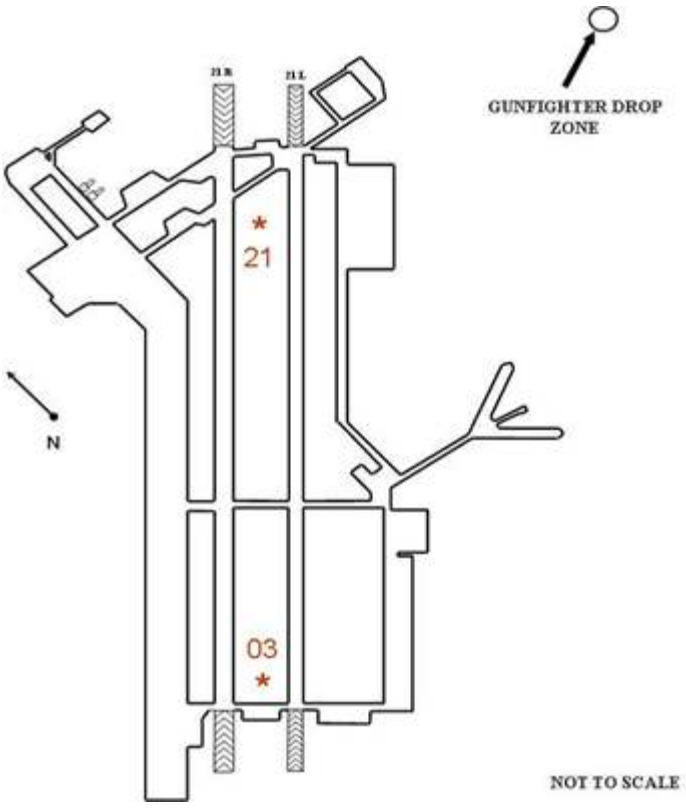
Weather Watch—Issued when the potential exists for severe weather. Organizations and personnel should consider taking protective measures.

World Meteorological Organization (WMO)—The specialized agency of the United Nations for meteorology (weather and climate), operational hydrology and related geophysical sciences; it has a membership of 189 countries and territories.

Attachment 2

LOCATION OF AN/FMQ-19

Figure A2.1. LOCATION OF AN/FMQ-19



Attachment 3

SPECIAL OBSERVATION CRITERIA

A3.1. General. Special observations (SPECI) are taken whenever specific criteria significant to both flight operations and resource protection occur. Criteria directing these observations are derived from guidance in AFMAN 15-111, *Surface Weather Observations*, and Flight Information Publications (FLIP). SPECI criteria will be updated in the form of a MFR in the event of corresponding FLIP updates that occur more frequently than this document is updated.

A3.2. Procedures: The FMQ-19 or duty forecaster (when augmenting) will take and disseminate a SPECI observation whenever any of the following criteria are observed. If the change occurs while taking an hourly observation, the criteria will be included in the scheduled hourly observation and labeled as a Record (METAR) observation:

Table A3.1. Thresholds for Ceiling, Visibility, and RVR Special Criteria.

The ceiling (rounded off to reportable values) forms or dissipates below, decreases to less than, or, if below, increases to equal or exceed the values listed below.	Surface visibility as reported in the body of the report decreases to less than, or, if below, increases to equal or exceed the values listed below.	Runway Visual Range (RVR) for active runway decreases to less than or, if below, increase to equal or exceed.
5,000 ft (FAA ATIS remark requirement)	**4 SM** (AFMAN 15-111) (TWR VIS criteria)	**6000 ft** (AFMAN 15-111)
3,000 ft (AFMAN 15-111)	**3 SM** (AFMAN 15-111)	**5000 ft** (AFMAN 15-111)
2,000 ft (AFMAN 15-111)	**2 1/2 SM** * (DoD Flips)	**4,500 ft** (DoD Flips)
1,500 ft (AFMAN 15-111)	**2 1/4 SM** (DoD Flips)	**4000 ft** (DoD Flips)
1,200 ft (DoD Flips)	**2 SM** (AFMAN 15-111)	**2400 ft** (AFMAN 15-111)
1,000 ft (AFMAN 15-111)	**1 3/4 SM** (DoD Flips)	**2000 ft** (AFMAN 15-111)
900 ft (DoD Flips)	**1 1/2 SM** (AFI 11-202v3 ACC SUP, Table 8.3, AFMAN 15-111)	**1800 ft** AFI 11-202v3 ACC SUP))
800 ft (AFMAN 15-111)	**1 3/8 SM** * (DoD Flips)	**1600 ft** (AFMAN 15-111)
700 ft (AFMAN 15-111)	**1 1/4 SM** (DoD Flips)	**1200 ft** (AFMAN 15-111)
600 ft (DoD Flips)	**1 SM** (AFMAN 15-111)	**1000 ft** (AFMAN 15-111)
500 ft (AFMAN 15-111)	***7/8 SM** (DoD Flips)	**600 ft** (AFMAN 15-111)
300 ft (AFMAN 15-111)	**¾ SM** (DoD Flips)	
200 ft (AFMAN 15-111)	**½ SM** (AFMAN 15-111)	RVR is first determined as unavailable (RVRNO) for runway in use, and when first determined that RVRNO report is no longer applicable, provided RVR conditions exist.
100 ft (AFMAN 15-111)	**¼ SM** (AFMAN 15-111)	RVR report required when prevailing visibility first

		observed < 1SM/1600 meters, again when prevailing visibility goes above 1SM/1600 meters
Note: Items marked with an (*) indicate the FMQ-19 will NOT sense this special criterion		

Table A3.2. Additional Special Criteria.

Criteria	Pertinent Data	Notes
Tower Visibility (1) Notified by the control tower that tower visibility has decreased to less than or, if below, increased to equal or exceed 1, 2, or 3 statute miles, AND the control tower visibility differs from the prevailing visibility. (2) Notified by the control tower that tower visibility has decreased to less than or, if below, increased to equal or exceed locally developed special criteria AND the control tower visibility differs from the prevailing visibility.	Transmit a SPECI with the tower visibility as a remark when the criteria to the left are met.	**Note 1**
Sky Condition An observation will be taken when a layer of clouds or obscuring phenomenon aloft is observed below the highest published instrument landing minimum (including circling) applicable to the airfield (1,100 feet), and no layer aloft was reported below this height in the previous METAR or SPECI observation.		
Wind Shift The wind direction changes by 45 degrees or more in less than 15 minutes with sustained winds (or gust) of 10 knots or more throughout the wind shift.		
Squall A strong wind characterized by a sudden onset in which the wind speed increases at least 16 knots and is sustained at 22 knots or more for at least one minute.	A SPECI is not required to report a squall if one is currently in progress.	
Volcanic Ash When first observed.		**Note 1**
Thunderstorm Begins or ends, i.e. no thunderstorm activity for 15 minutes.	A special observation is not required to report the beginning of a new thunderstorm if one is currently reported in progress at the weather station.	
Precipitation (1) Hail begins or ends	Except for freezing rain, freezing drizzle, hail and ice pellets, a	**Note 2**

(2) Freezing precipitation begins, ends or changes intensity. (3) Ice pellets begin, end or change intensity (4) Any other type of precipitation begins or ends (e.g., rain, drizzle, snow, etc.)	SPECI is not required for changes in type (e.g., drizzle changing to snow grains) or the beginning or ending of one type while another is in progress (e.g., snow changing to rain and snow).	
Tornado, Funnel Cloud or Waterspout When first observed or disappears from sight or ends.		**Note 1**
Upon Resumption of Observing Function (1) Within 15 minutes after returning to duty following a break in observing coverage or augmentation at the observing location unless a record observation is filed during that 15-minute period. (2) Within 15 minutes of establishing operations in the alternate observing site if FBWOS is not functioning.		
Aircraft Mishap * Not disseminated. * A SPECI observation is not required for in-flight emergencies; however, these emergencies alert weather personnel to intensify the weather watch.	During normal weather flight hours of operation only	
Note 1: The FMQ-19 will NOT sense this special criterion Note 2: FMQ-19 will not sense hail or ice pellets.		

Attachment 4

TAF SPECIFICATION AND AMENDMENT CRITERIA

A4.1. General. When certain weather thresholds (ceiling, visibility, winds, precipitation, etc.) are met, the change will be reflected in the forecast for the time of occurrence (nearest hour), duration and intensity. These thresholds are called specification criteria. **Table A4.1** lists specification and amendment criteria for Nellis AFB.

Table A4.1. Standard Specification and Amendment Criteria

Ceilings	Visibility	Category
GTE 2000 FT	GTE 3 SM (4800 M)	**E**
LT 2000 FT but GTE 1000 FT	LT 3 SM (4800 M) but GTE 2 SM (3200 M)	**D**
LT 1000 FT but GTE 700 FT	LT 3 SM (4800 M) but GTE 2 SM (3200 M)	**C**
LT 700 FT but GTE 200 FT	LT 2 SM (3200 M) but GTE 1/2 (0800 M)	**B**
LT 200 FT	LT 1/2 SM (0800 M)	**A**
Surface Wind Speeds	Specify wind speed changes of 10 knots or more.	Amend if predominant wind speed is in error by 10 knots or more.
Surface Wind Gusts	Specify onset, duration, and intensity of wind gusts.	Amend if wind gust speed is in error by 10 knots or more.
Surface Wind Direction	Specify a change in prevailing wind direction of more than 30 degrees when the predominant wind speed or gusts are expected to be 15 knots or more.	Amend if prevailing wind speed is in error by more than 30 degrees AND winds are 15 knots or more.
Icing	Specify Icing not associated with thunderstorms from surface to 10,000ft AGL.	Amend if beginning or ending of icing meets, exceeds, or decreases to less than or greater than moderate intensity and was not specified in the forecast.
Turbulence	(Weather Category II Aircraft) Specify turbulence not associated	Amend if the beginning or ending of turbulence meets, exceeds, or

	with thunderstorms from surface to 10,000 ft AGL.	decreases to less than moderate or greater intensity and was not specified in the forecast.
Weather Warning Criteria and/or Weather Advisory Criteria	Specify the onset, duration, and intensity of weather warning criteria.	Including non-convective low-level wind shear: Occur, or are expected to occur, during the forecast period, but were not specified in the forecast. Were specified in the forecast, but are no longer occurring or expected to occur during the forecast period.
Altimeter Setting		Altimeter setting meets or exceeds 31.00 INS and was not specified in the forecast; if above, drops below 31.00 INS and was not specified during the forecast period; drops below 28.00 INS and was not specified in the forecast; or if below 28.00 INS, increases above 28.00 INS and was not specified in the forecast.
Thunderstorms	Specify onset and duration of thunderstorms at the aerodrome complex.	Amend if the start or end time of the thunderstorm is incorrectly specified.
Temporary Conditions	Specify the onset and duration of temporary conditions.	1. Amend if temporary conditions become predominant. 2. Amend if temporary conditions do not occur as forecast. 3. Amend if temporary conditions are no longer expected to occur.
Changes to Predominant Conditions	Specify the onset, duration, and intensity (if applicable) of changes to predominant conditions.	Forecast change conditions (BECMG or FM group) occur before the beginning of the specified period of change and are expected to persist; do not occur within 30

		minutes after the specified time; or are no longer expected to occur.
Representative Conditions		Amend if forecast conditions are considered unrepresentative of existing or forecast conditions and amending the forecast improves safety, flight planning, operations efficiency, or assistance to in-flight aircraft.

Attachment 5

MISSION WEATHER THRESHOLDS / AIRCRAFT SENSITIVITIES

A5.1. General MEFs are verified using criteria based on airframe and mission type. Due to the myriad of mission types flown by aircraft at Nellis AFB, the weather flight classifies missions types into nine separate categories.

Table A5.1. USAF General Flight Rules AFI11-202V3 4.15.3.

Aircraft Type	Weather Condition	Impact
Fixed Wing Aircraft	CIG <=2,000ft or VIS <=3SM	Require Alternate
Helicopters	CIG<=1,000ft (or 400ft above lowest approach minimums) or VIS <=2SM	Require Alternate
NOTE: Takeoff and Landing Restrictions per Flight Information Publications (FLIPs), High and Low Altitude, unless stated otherwise below. FLIPs are reviewed quarterly.		

Table A5.2. Cig/Vis – Take-off/Landing Restrictions.

Aircraft Type	Ceiling	SFC Visibility
HH-60	≤ 1,000ft (Unaided) < 700 (NVG) ≤ 700ft (Day)	< 3 SM (Unaided) < 2 SM(NVG) < 1 SM(Day)

Table A5.3. Crosswinds. Derived from local flight checklists, AF11-2F series (aircraft specific) and TTPs.

Aircraft Type	Cross Winds/Dry	Cross Winds/Wet
A-10	> 35KT	*
F-35	> 25KT	> 20KT
F-15C	> 30KT	*
F-15E	> 30KT	*
F-16	> 25KT	> 23KT
F-22A	> 30KT	*
HH-60	*	*

Table A5.4. Hazardous Weather – Enroute Hazards – Unless stated otherwise, restricts mission operations

Aircraft Type	Turbulence	Icing	WX
A-10	Observed MDT	Forecast MDT	Avoid all TS
F-35	*	*	Avoid by >25nm
F-15C	Forecast MDT	Forecast LGT, may penetrate	Avoid, if psbl Avoid by 10nm
F-15E	Avoid MDT-SVR	May penetrate, avoid SVR	<FL250
F-16	Avoid obs SVR	Avoid MDT	Avoid, if psbl
F-22A	*	LGT	Avoid
HH-60	MDT	MDT	Avoid

Table A5.5. Mission Restrictions.

Mission Type	OP Area Ceiling (Day)	OP Area Ceiling (Night)
AA	100 AGL	100 AGL
AIRDEMO	015 AGL	015 AGL
CAS	015 AGL	050 AGL
CSAR	007 AGL	007 AGL
DCS	007 AGL	No Criteria
LOWAT	050 AGL	100 AGL
ME	007 AGL	015 AGL
NTF	No Criteria	008 AGL
SA	100 AGL	100 AGL

Mission Type	Surface Visibility (Day)	Surface Visibility (Night)	Flight Level Visibility
AA	< 5 SM	< 5 SM	No Criteria
AIRDEMO	< 5 SM	< 5 SM	No Criteria
CAS	< 3 SM	< 5 SM	< 3 NM
CSAR	< 1 SM	< 2 SM	No Criteria
DCS	< 1 SM	< 1 SM	No Criteria
LOWAT	No Criteria	No Criteria	No Criteria
ME	< 5 SM	< 5 SM	No Criteria
NTF	< 2 SM	< 2 SM	No Criteria
SA	< 5 SM	< 5 SM	No Criteria

Mission Type	TSTMS in Op Area	Range Wind	Turbulence	Icing
AA	25% Coverage	>= 35 KT	No Criteria	Severe
AIRDEMO	25% Coverage	>= 35 KT	No Criteria	Severe
CAS	25% Coverage	>= 35 KT	No Criteria	Severe
CSAR	25% Coverage	>= 45 KT	No Criteria	Moderate
DCS	25% Coverage	>= 40 KT	No Criteria	Light
LOWAT	25% Coverage	>= 35 KT	No Criteria	Severe

ME	25% Coverage	>= 35 KT	No Criteria	Light
NTF	25% Coverage	>= 40 KT	No Criteria	Light
SA	25% Coverage	>= 35 KT	No Criteria	Severe

Attachment 6

MISSION EXECUTION FORECAST (MEF) AMENDMENT CRITERIA

A6.1. General. Table A6.1 lists the amendment criteria to be applied to the Take-Off and Landing portions of the MEF. Table A6.2 lists the amendment criteria to be applied to the Range Weather Forecasts section of the MEF and is based on the Mission Limiting Parameters outlined in Attachment 5.

Table A6.1. MEF Amendment Criteria for Takeoff and Landing Data.

Item		Remarks
Ceilings	≥ 2000 FT < 2000 FT but ≥ 1000 FT < 1000 FT but ≥ 700 FT < 700 FT but ≥ 200 FT < 200 FT	Amend if ceilings are forecast to cross, or are observed at the following thresholds.
Visibility	≥ 3 SM (4800 M) < 3 SM (4800 M) but ≥ 2 SM (3200 M) < 2 SM (3200 M) but ≥ 1/2 SM (0800 M) LT 1/2 SM (0800 M)	
Surface Wind Speeds	Specify wind speed changes of 10 knots or more	Amend if predominant wind speed is in error by 10 knots or more
Surface Wind Gusts	Specify onset, duration, and intensity of wind gusts	Amend if wind gust speed is in error by 10 knots or more
Surface Wind Direction	Specify a change in prevailing wind direction of more than 30 degrees when the predominant wind speed or gusts are expected to be 15 knots or more	Amend if prevailing wind speed is in error by more than 30 degrees AND winds are 15 knots or more
Icing	Specify Icing not associated with thunderstorms from surface to 10,000ft AGL	Amend if beginning or ending of icing meets, exceeds, or decreases to less than or greater than moderate intensity and was not specified in the forecast
Turbulence	(Weather Category II Aircraft) Specify turbulence not associated with thunderstorms from surface to 10,000 ft AGL	Amend if the beginning or ending of turbulence meets, exceeds, or decreases to less than moderate or greater intensity and was not specified in the forecast
Weather Warning Criteria and/or Weather Advisory Criteria	Specify the onset, duration, and intensity of weather warning criteria	Including non-convective low-level wind shear: Occur, or are expected to occur, during the forecast period, but were not specified in the forecast. Were specified in the forecast, but are no longer occurring or expected to occur during the forecast period

Altimeter Setting		Altimeter setting meets or exceeds 31.00 INS and was not specified in the forecast; if above, drops below 31.00 INS and was not specified during the forecast period; drops below 28.00 INS and was not specified in the forecast; or if below 28.00 INS, increases above 28.00 INS and was not specified in the forecast.
Thunderstorms	Specify onset and duration of thunderstorms at the aerodrome complex	Amend if the start or end time of the thunderstorm is incorrectly specified
Temporary Conditions	Specify the onset and duration of temporary conditions	1. Amend if temporary conditions become predominant. 2. Amend if temporary conditions do not occur as forecast. 3. Amend if temporary conditions are no longer expected to occur.
Changes to Predominant Conditions	Specify the onset, duration, and intensity (if applicable) of changes to predominant conditions	Forecast change conditions (BECMG or FM group) occur before the beginning of the specified period of change and are expected to persist; do not occur within 30 minutes after the specified time; or are no longer expected to occur.
Representative Conditions		Amend if Forecast conditions are considered unrepresentative of existing or forecast conditions and amending the forecast improves safety, flight planning, operations efficiency, or assistance to in-flight aircraft. E.g. Altimeter Setting

Table A6.2. MEF Amendment Criteria for Range Weather Forecast.

Item	Remarks
Ceilings	Unforecasted change in ceilings resulting in the operating area ceiling crossing one of the following levels (AGL): 14,000 ft; 10,000 ft; 5,000 ft; 1,500 ft; 800 ft (Night); 700 ft (Day)
Airspace Sky Condition	Unforecasted change in sky condition in the airspace crossing one of the following thresholds (MSL): 10,000 ft clear airspace between 10K – 30K; Clear below 10,000 ft; (MSL) Clear below 2,000 ft (AGL)
Surface Visibility	Unforecasted change in visibility in the airspace crossing one of the following thresholds: 5 SM (8000 M); 3 SM (4800 M); 2 SM (3200 M); 1 SM (1600 M)
Surface Winds	Unforecasted change in surface winds resulting in wind speeds, sustained or gust, crossing the following thresholds: 40KT; 35KT
In-Flight Hazards	Unforecasted change of turbulence intensity crossing through the moderate category for the airframe in the operating area. Unforecasted change of icing intensity crossing through the light category in the operating area. Unforecasted change start/stop of thunderstorms in the operating area.
Issuance/Cancellation/Extension of a weather watch, warning or advisory	

Attachment 7

WEATHER WATCH, WARNING, AND ADVISORY (WWA) CRITERIA

A7.1. General. Table A7.1 lists all WWA's issued for Nellis AFB and the NTTR. The following is listed: WWA type, criteria of the product issued, and the desired lead time (DLT). With the exception of advisories/watches in bold, all products are issued for a 5 nm radius from the center point of the airfield at Nellis AFB.

Table A7.1. Weather Watch, Warning, and Advisory Criteria.

WWA Type	Criteria	DLT	Impact	INWS
Warning	Damaging Winds >= 50 kts. at Nellis AFB, not associated with convection (PHASE 3 / SWAP)	1:30	Hangar Aircraft according to MXG/CC and NAFB 011. Consider canceling/diverting aircraft operations. Close all canopies, and instruct personnel to seek shelter, either indoors or under aircraft.Shelter all vehicles and equipment subject to damage	yes
Warning	Freezing Precipitation expected at Nellis AFB (SWAP)	1:00	Poses significant risk of damage to facilities and rapidly creates hazardous conditions for personnel and vehicle movement that cannot easily be mitigated	yes
Warning	Heavy Snow >= 2 inches. within 12 hrs. expected at Nellis AFB	1:00	Base Protection IAW NAFB Plan 10-2	yes
Warning	Observed Lightning within 5nm occurring at Nellis AFB and Area 2	0	Cease flight line operations and munitions operations	yes
Warning	Heavy Rain >= 1/2 inches. within 1 hrs. Flash Flooding expected and imminent at Nellis AFB	1:00	Increased threat of flash flooding or systemic flooding posing credible threat to unprotected resources and personnel Disrupts flight line and maneuver activities Imposes increased risk on personnel movement	yes
Warning	Strong Winds >= 35 but < 50 kts. at Nellis AFB. Not associated with convection (PHASE 2)	1:30	Mooring Aircraft for A-10 and HH-60s. Remove or tie down maintenance	yes

			stands on the flight line. Ensure hangar doors are closed.	
Warning	Tornado expected at Nellis AFB (SWAP)	:30	Secure as much equipment as time permits. Terminate all outside activities and direct personnel to take shelter immediately Consider canceling/diverting aircraft operations.	yes
Warning	Moderate Thunderstorms (SWAP). High Wind >= 35 but < 50 kts. and/or Large Hail >= 1/4 but < 3/4 inch.	1:30	Hangar Aircraft according to MXG/CC and NAFB 011. Consider canceling/diverting aircraft operations. Close all canopies, and instruct personnel to seek shelter, either indoors or under aircraft.Shelter all vehicles and equipment subject to damage	yes
Warning	Severe Thunderstorms (SWAP). Damaging Hail >= 3/4 inch. and/or Damaging Wind >= 50 kts.	1:30	Hangar Aircraft according to MXG/CC and NAFB 011. Consider canceling/diverting aircraft operations. Close all canopies, and instruct personnel to seek shelter, either indoors or under aircraft.Shelter all vehicles and equipment subject to damage	yes
Watch	Potential for Damaging Winds >= 50 kts. exists at Nellis AFB (PHASE 3 / SWAP)	3:00		yes
Watch	Potential for Freezing Precipitation exists at Nellis AFB (SWAP)	3:00		yes
Watch	Potential for Heavy Rain >= 1/2 inches. within 1 hrs. with increased potential of Flash Flooding exists at Nellis AFB	3:00		yes
Watch	Potential for Heavy Snow >= 2 inches. within 12 hrs. exists at Nellis AFB	3:00		yes

Watch	**Potential for Lightning within 15 NM. of Nellis AFB**	**1:00**	Hangar F-35/Download Munitions	no
Watch	Potential for Lightning within 5 NM. at Nellis AFB and Area 2	:30		yes
Watch	Potential for Strong Winds >= 35 but < 50 kts. at Nellis AFB (PHASE 2)	3:00		no
Watch	Potential for Tornadic Activity exists at Nellis AFB (SWAP)	1:00		yes
Watch	Potential for Moderate Thunderstorms. Large Hail >= 1/4 but less than 3/4 inch. and/or Strong Winds >= 35 but less than 45 knots.	3:00		no
Watch	Potential for Severe Thunderstorms (SWAP). Damaging Hail >= 3/4 inch. and/or Damaging Wind >= 50 kts.	3:00		yes
Advisory	**Observed Any Intensity icing (SFC-100) not associated with thunderstorms within 10NM of Nellis AFB**	**0**		no
Advisory	Observed Crosswinds greater than or equal to 20 knots.	0	Decision to cancel flying or divert aircraft (F-35)	no
Advisory	Observed Crosswinds greater than or equal to 25 kts.	0	Decision to cancel flying or divert aircraft	no
Advisory	Observed ICE FOD Temperature <= 7 and Dewpoint Depression <= 5 C.	0	Cover or monitor engine inlets (F-16) for ice accumulation	no
Advisory	**Observed Lightning/Thunderstorms within 15 NM. occurring**	**0**	**Hangar F-35/Download Munitions and cease flightline and munitions operations**	**no**
Advisory	**Observed Lightning/Thunderstorms observed within 25 NM.**	**0**	**Cease F-35 Operations/Download Munitions**	**no**
Advisory	**Observed Moderate or Greater Turbulence is occurring within 10 NM. of Nellis AFB from SFC-10,000ft**	**0**		**no**
Advisory	Forecast Surface Winds >= 25 but < 35 kts. (PHASE 1)	:30	Prepare to hangar light aircraft. If space is unavailable, remaining aircraft will be tied	no

			down and all wheels chocked. Terminate all non-essential movement of munitions. Remove all fire extinguishers from the flight line.	
Advisory	**Observed Sustained Surface Winds of >= 35KT within NTTR**	**0**		**no**
Advisory	Observed Fighter Thermal Stress Danger occurring	0	Determines amount of time pilots can safely work outside their aircraft.	no
Advisory	Observed Fighter Thermal Stress Caution occurring	0	Determines amount of time pilots can safely work outside their aircraft.	no
	* This criteria is issued only during Wing and Exercise Flying*			

A7.2. General. The dissemination of National Weather Service (NWS) warnings (river flood, flash flood, fire weather (red flag), coastal flood, and tsunami watches) has been coordinated through the Nellis AFB Emergency Management (EM) Flight per AFMAN 15-129. The EM Flight has determined that none of these criteria are required to be issued for Nellis AFB and its surrounding areas.